Formalizing Thirty-Three Miniatures

Implementing Applications of Linear Algebra in Isabelle/HOL

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Formalization Goals

Formalizing Mathematics

- Formalize proof strategies
- Focus on linear algebra methods
- Build on **TODO book**
- Benefits of proof assistants

Exploring Libraries

- Build on existing formalizations
- Archive of Formal Proofs + Isabelle/HOL
- Extend linear algebra libraries

Applying Isabelle/HOL

- Understand Isabelle features
- Write reusable proofs

Formalizing Mathematics: Thirty-Three Miniatures

Miniature 2

Formalizing Mathematics: Thirty-Three Miniatures

Miniature 3

Formalizing Mathematics: Thirty-Three Miniatures

Thirty-Three Miniatures: Miniature 5

Formalizing Mathematics: Learnings

Theorem Provers Encourage Abstraction

Abstraction makes proofs less verbose (less options for try, searching for relevant lemmas by hand is easier)

Concrete Examples Can Help (#instantiateYourLocales)

Concrete application of definitions avoids inconsistencies and hopeless proof attempts

- First oddtown definition did not require sets to be different in the oddtown predicate -> empty set, easy to prove?
- Hamming Code definition issues: universal quantification over code words instead of all words -> proof broke?

Algebra Libraries in Isabelle/HOL

- Algebraic structures as type classes (carrier = type universe) vs. locales (carrier explicitly given) [mention typedef, Rep and Abs]
- using ring-scheme for objects that do not have a multiplication

Algebra Libraries in the Archive of Formal Proofs

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Useful Isabelle/HOL Features

- Commands: proof-cases, try
- Types: records, typedef
- Update syntax: function updates (see Nitpick), record updates (subspace.vs)
- Locales (“locale1 + ...” leads to automatic instantiations of locale1-parameters and directly yields the assumptions of locale1)
- Proof structure: blocks (idk, the $\{\ldots\}$ thingies) (according to some people this makes proofs less understandable so maybe not really that useful)

- meme: isabelle logo

Eisbach

content... (TODO (cope))

Weird Stuff

- leaving out "and"s in assumes/shows clauses changes the behaviour (leaving out the "shows" before multiple goals of a lemma leads to "thm" only referencing the first goal instead of all of them)
- lemmas that are unexpectedly not found by try0 (e.g. $n > 0 \Rightarrow \exists m.n = Suc\ m$)
- no error on forgotten "end" after theory/context
- locale instantiations with fixed parameters with no assumptions curried?
- "?" behind added locale 'name' makes 'name.def' optional (approved)
- list comprehension $[1 :: nat..3 :: nat]$ becomes *int list* but $[1 :: nat..< 3 :: nat]$ gives *nat list*
- Non-aligning AFP and Isabelle version
- termination proofs for non-recursive functions (defined by "function") optional but if not provided, simps rules are whack
- termination using 'termination' by blast????